

Question

Rhodium(III) chloride trihydrate ($\text{RhCl}_3 \cdot 3\text{H}_2\text{O}$) is a red solid, but its composition is relatively complex; column chromatographic separation and spectroscopic studies of its aqueous solution reveal the presence of two hexacoordinate mononuclear complex molecules, **A**₁, **A**₂; in **A**₁, the chemical environment of the same kind of atom is the same. The reaction of $\text{RhCl}_3 \cdot 3\text{H}_2\text{O}$ with CO gas flow yields **B**; **B** is a dinuclear complex molecule, Rh is 4-coordinate, +1 oxidation state, and the ligands are only Cl^- and CO. **B** reacts with tetramethylammonium iodide to obtain **C**; **C** is a 1:1 type salt, Rh is in the +1 oxidation state, and the ligands are only I^- and CO. **B** is a diamagnetic species.

Using the anion of **C** as a catalyst and iodomethane as a co-catalyst, the preparation of acetic acid can be achieved: $\text{CH}_3\text{OH} + \text{CO} \rightarrow \text{CH}_3\text{COOH}$. In the reaction, the anion of **C** undergoes an oxidative addition reaction with iodomethane to obtain **D**; **D** undergoes methyl migration to obtain **E**; **E** coordinates with CO to obtain **F**; **F** undergoes reductive elimination, releasing the small molecule **G** and regenerating **C**; **G** reacts with water to produce acetic acid and HI; HI reacts with methanol to produce iodomethane and water.

There are the following statements:

- 1, Ignoring hydrogen atoms, the point groups to which **A**₁, **A**₂, **B** belong are C_{2v} , C_{3v} , D_{2h} , respectively.
- 2, The theoretical magnetic moments of **A**₁, **C** are 0 and 2.83 B.M., respectively.
- 3, The oxidation states of Rh in **D**, **E**, **F** are not all the same.
- 4, The molecular weight of **G** is 297.9.

Then the option that contains all the correct statements is:

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1, 2

G. 1, 3

H. 1, 4

I. 2, 3

J. 2, 4

K. 3, 4

L. 1, 2, 3

M. 1, 2, 4

N. 1, 3, 4

O. 2, 3, 4

P. 1, 2, 3, 4

Answer

Correct Answer: **A**

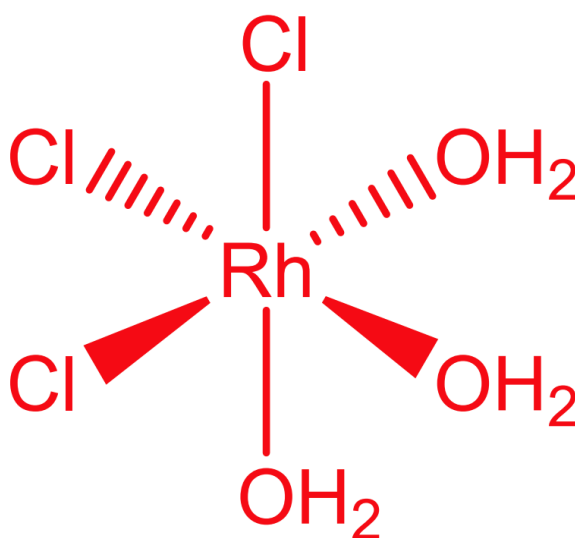
Detailed Explanation

$\text{RhCl}_3 \cdot 3\text{H}_2\text{O}$ exists in both mer and fac isomers, where the fac isomer has the same chemical environment for the same type of atoms. Therefore, **A₁** is the fac isomer, and **A₂** is the mer isomer.

CHECKPOINT

1 PTS

A₁ is the

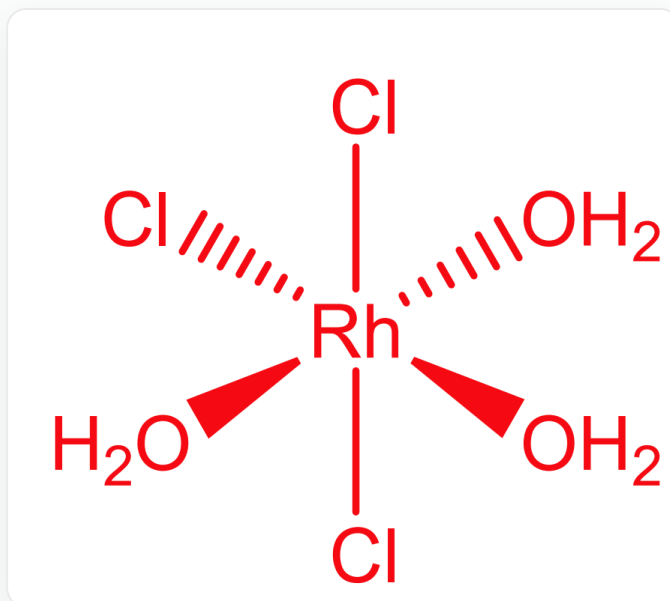


fac structure of $\text{RhCl}_3 \cdot 3\text{H}_2\text{O}$

CHECKPOINT

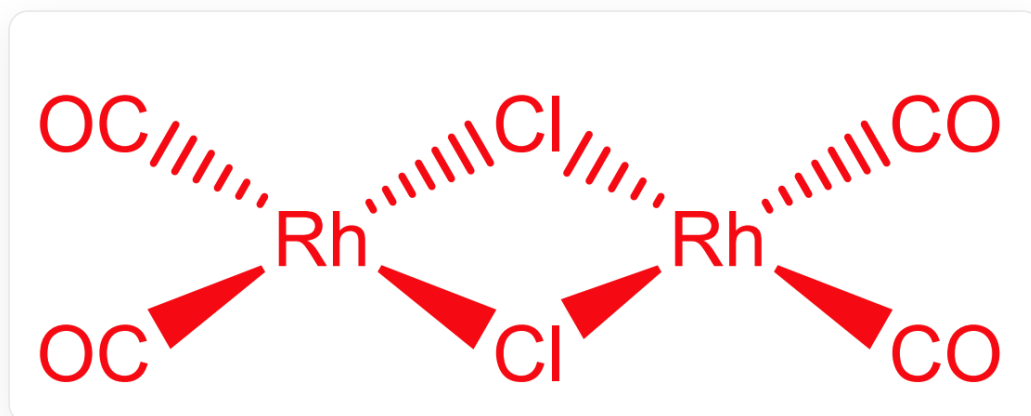
1 PTS

A₂ is the



mer structure of $\text{RhCl}_3 \cdot 3\text{H}_2\text{O}$

The reaction of $\text{RhCl}_3 \cdot 3\text{H}_2\text{O}$ with CO gas flow yields **B**, which is a dinuclear complex. Rh is 4-coordinated, in a +1 oxidation state, and **B** is a diamagnetic species; therefore, the coordination geometry is square planar. Thus, the structure of **B** can be deduced as

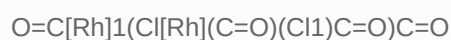
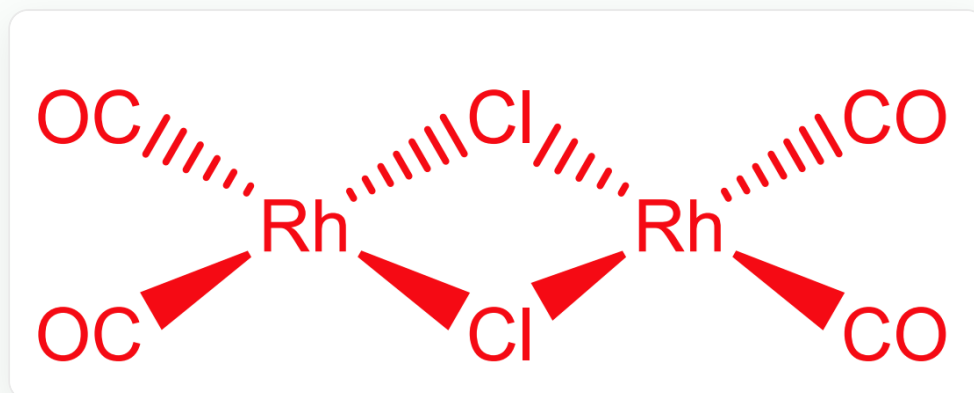


$\text{O}=\text{C}[\text{Rh}]_1(\text{Cl}[\text{Rh}])(\text{C}=\text{O})(\text{Cl})\text{C}=\text{O}$

CHECKPOINT

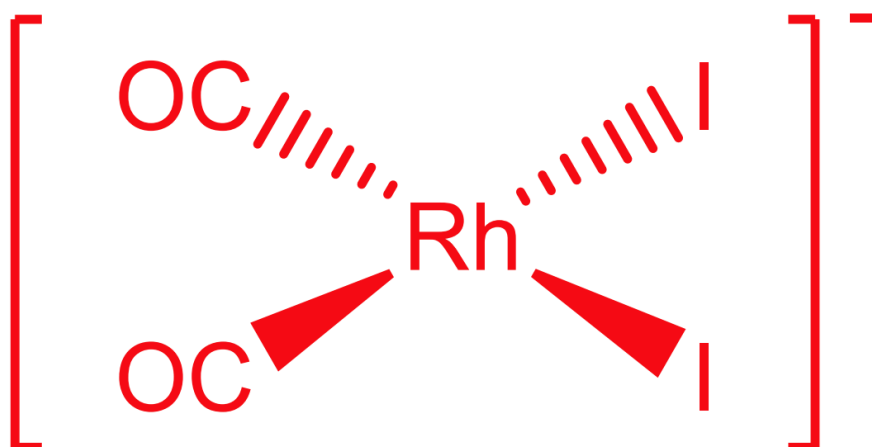
2 PTS

B is



Therefore, the point groups to which **A**₁, **A**₂, **B** belong are C_{3v} , C_{2v} , D_{2h} respectively. Statement 1 is incorrect.

B reacts with tetramethylammonium iodide to yield **C**; **C** is a 1:1 type salt, Rh is in the +1 oxidation state, and the ligands are only I^- and CO. Therefore, there are two I^- and two CO ligands. Based on the positional relationship of CO in **B**, the structure of **C** can be obtained as



$[Rh-](I)([C]=O)[C]=O$, the two CO groups are in cis position

CHECKPOINT

1 PTS

C does not have any unpaired electrons

The theoretical magnetic moments of **A**₁, **C** are both 0. Statement 2 is incorrect.

C undergoes oxidative addition with iodomethane, readily yielding **D** with the chemical formula $[Rh(CO)_2I_3(CH_3)]^-$. **D** undergoes methyl migration to yield $[Rh(CO)I_3(CH_3CO)]^-$; $[Rh(CO)I_3(CH_3CO)]^-$ coordinates with CO to yield $[Rh(CO)_2I_3(CH_3CO)]^-$; $[Rh(CO)_2I_3(CH_3CO)]^-$ undergoes reductive elimination, eliminating the small molecule CH_3COI and regenerating **C**.

CHECKPOINT

3 PTS

D is $[Rh(CO)_2I_3(CH_3)]^-$, **E** is $[Rh(CO)I_3(CH_3CO)]^-$, **F** is $[Rh(CO)_2I_3(CH_3CO)]^-$

CHECKPOINT

1 PTS

G is CH_3COI

The oxidation state of Rh in **D, E, F** is all +3, and the molecular weight of **G** is 169.9. Statements 3 and 4 are incorrect.

Choose A